

# Photometric distance and age of the open cluster NGC 2175

Alec Stanley and Jessel Jolly\*

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## Abstract

We determine the distance and age of the open cluster NGC 2175. This was completed using R, V, and B band observations from a SeeStar telescope obtained on March 21, 2026. To determine the distance we overlaid the data on a set of calibrated isochrones obtained from the CMD 3.9 data set, maintained by Léo Girardi at the Osservatorio Astronomico di Padova, to get a distance of 1.5 kpc. By fitting our data to these constant-age isochrones we also determine the cluster to have an age of 2-3.16 Myr. NGC 2175 is known to be a young open cluster, and these findings agree with literature.

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\*Student IDs: 33963975 and 34820418 respectively

# 1 Introduction

The open cluster NGC 2175 lies within the Monkey Head Nebula (NGC 2174) and forms part of the Orion constellation. This cluster and nebula were first identified as an HII region by Sharpless in 1959, originally being catalogued as S252 (Chibueze et al. 2013). It is known to be at a distance of  $2.2 \pm 0.3$  kpc (Khachikian et al. 2005). Other higher estimates give 2.6 kpc by Grasdalen and Carrasco 1975. The age of the open cluster was estimated by Khachikian et al. 2005 to be 2 Myrs, an extremely young cluster dominated by main and pre-main sequence stars.

In this report we aim to use modern methods of computational aperture photometry to form our own independent results for the age and distance of open cluster NGC 2175. This was done using a SeeStar telescope in 2026 which was processed to obtain R, V, and B band magnitudes. We will discuss the data we obtained, the processing, and our final measurements below.

## 2 Method

In order to measure the properties of NGC2175, we used a SeeStar S50 telescope on the 21st March 2026. This telescope is a portable, self-guided telescope with a 50mm aperture which was used at the Hutton-Westfold Observatory located next to the Monash University Clayton Campus. The SeeStar S50 used a Bayer matrix in order to sample light in blue, green and red spectra. The in-built software of the SeeStar S50 is designed to pre-process the image. This includes accounting for the affects of the flat, bias and dark current images. Thereby, in the processing steps, these will not need to be accounted for.

During the data collection process, an issue arose with the telescope pointing closer and closer to a tree over time. As this occurred, light scattering off the tree entered the telescope causing a large section of the images to artificially be brighter. Once this was noticed, we moved the telescope to a different location where this was not an issue, however, many of our images were of very poor quality. A significant trend due to this scattered light was the total flux of the image would continuously increase over time. In order to remove the effected images, we normalised the total fluxes to that of an image of good quality, and removed images which had a normalised total flux of 1.2 or greater. This line was determined as it produced the best quality images and removed the majority of low quality images. The signal to noise ratio depends on the number of images in Poisson statistics by,

$$\text{SNR} \propto \sqrt{N}. \quad (1)$$

Originally we had 290 images, after the removals, it became 149. Using Equation 1, we can find that our signal to noise ratio has reduced by 28%. While this is significant, compared to the low qualities of the removed images, this loss is necessary.

### 2.1 Alignment, Scaling, and Combining

The first step in processing the images was to align them all to the same position. As the Earth rotates, NGC 2175 physically shifts across the sky. While our telescope did a good job of following the object, we need to make sure that the translation and rotation of each image was perfectly aligned. We also had to move the telescope during our data collection which introduced a very significant shift. This process is a complicated series of translations and rotations which was done automatically using the ccdproc module. We shifted all images to align with an arbitrary reference image, chosen such that it avoided most of our bad data. This same reference image was used for each of the three filters.

As each image will be under slightly different conditions when taken, the fluxes may change as well. In order to combine images later, we must first scale the fluxes to be on the same scale. Across all the images, when taking the flux of a given star, we would expect it to be the same but it is not necessarily the case. Thereby, we will bring all images to the same scale by equating the fluxes of stars. To do this, we will choose a good quality image with little over-saturation and scale other images to fit the scale of this image. On each image, we took the fluxes of the same four stars and found the scale factor for each star to the target image

by dividing the fluxes of each star respectively. We then chose the median scale factor to account random variation in fluxes and then multiplied the counts of the entire image by this scale factor.

After alignment and scaling, our images are all in the same pixel grid and scale. Using this, we can combine image into a single image for each colour. This will be done by setting each pixel value to that of the median of the pixel value across all images, this will again be done using the `ccdproc` module. We use a median combine rather than mean combine as it is better at removing outliers from the image, such as hot pixels. Later for analysing, we will want to only focus on NGC 2175 and not everything surrounding it. We also have artifacts left over from the shifting process. For both these reasons, we chose to crop the images to include our object of interest, the Monkey Head Nebula with NGC 2175 within it.

Background noise is still present within the image. To facilitate the background subtraction, we segmented the image into 100 pixel squares. In each square, we can find the median background using the `photutils` Python module. This smoothly subtracts the image background, accounting for possible background changes across the images on a macroscopic scale. After all the processing steps, we have a single image in each of the three colour filters, red, blue and green. The final image in the red filter can be seen in Figure 1.

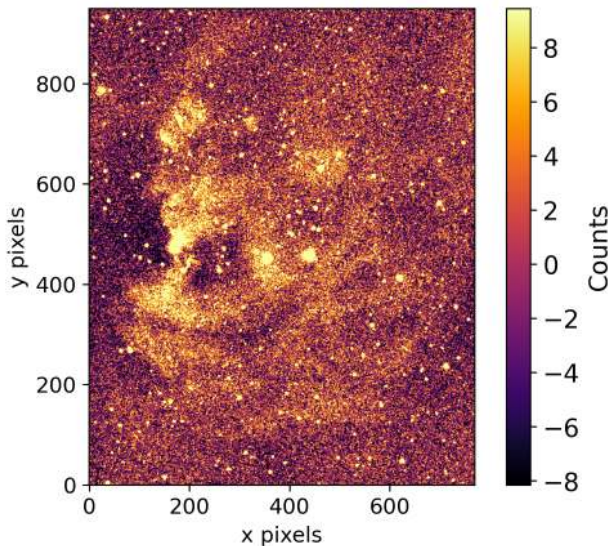


Figure 1: Final processed image of NGC 2175 in the red filter.

## 2.2 Cataloguing

The goal is to obtain a set of objects in our image, perform an analysis, and obtain measurements such as the distance and radius of the open cluster. The first step in such a process is using an algorithm that scans the flux of our images and determines, based on the image standard deviation, whether or not a bright spot in the image is an actual object or not. The tool we used for this was the source detection algorithm supplied by the `photutils` module. The immediate output of this process is shown in Figure 2.

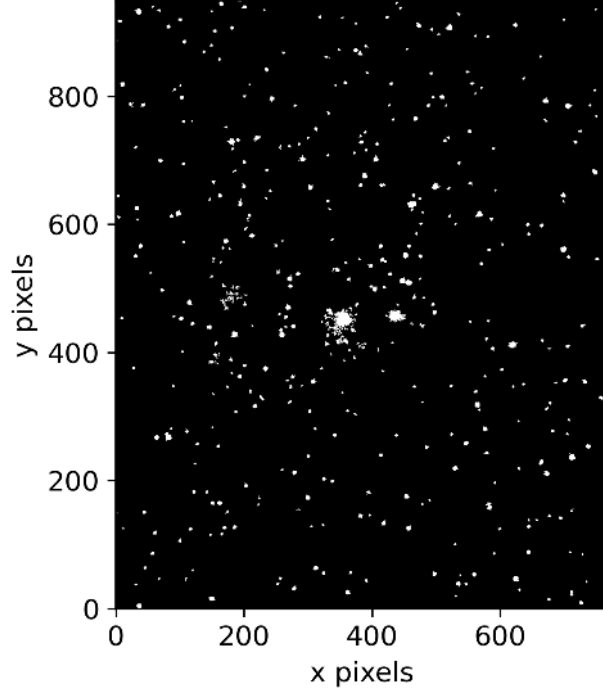


Figure 2: Segmented image displaying the sources in the green filter.

To be able to plot all our objects on a unified HR diagram, we need to obtain a set of calibrated magnitudes of all our objects. It is standard to first define the instrumental magnitude of a bright object as

$$m = -2.5 \log_{10}(f), \quad (2)$$

where the given flux  $f$  from the source image can be converted to an apparent magnitude, where lower is brighter. This is, however, not yet useful since we do not know how the instrumental magnitudes compare to standard magnitudes for objects in the sky, such as those used by Gaia for example. Our goal is to transform these instrumental magnitudes to a set of standard UBVRI magnitudes, and then using the colour measurements to obtain calibrated SeeStar magnitudes.

The set of Graham magnitudes of Gaia stars was obtained (Brown 2026) and used to determine the colours of hundreds of stars in the catalogue. We then obtained fit lines for each of these relations. We obtained  $V - G$ ,  $R - G$  and  $B - G$  all as a function of colour.

We then chose six unique calibration stars. We chose stars such that their magnitudes sat in the 9-12 range and were bright stars in our catalogue but not fully saturated. From these stars, median magnitudes can be obtained and can be directly substituted into our fit line equations empirically determined from the known catalogue.

The final SeeStar calibration equations were derived by Jamie Lawson and supplied by Michael Brown (Monash University) for the purposes of this report.

$$\text{SeeStar B} = B - 0.63(B - V) + 0.01 \quad (3)$$

$$\text{SeeStar G} = V - 0.07(B - V) + 0.01 \quad (4)$$

$$\text{SeeStar R} = R + 0.20(V - R) + 0.02 \quad (5)$$

After applying these equations, we have a set of SeeStar calibration magnitudes. By then subtracting the instrumental magnitudes from these, we obtain the zero points required to easily find the SeeStar magnitudes of any star in the catalogue. This was done by adding the zero point in each respective filter to the instrumental magnitude of each star.

### 3 Results and Analysis

To determine the distance to the open cluster, a common method is by comparing our data to isochrones. An isochrone is a line of stars on an HR diagram that have constant age. If we plot enough isochrones, we would be able to overlay our data on the same plot and experimentally determine a distance modulus and reddening result by shifting our data such that it aligns with the isochrones. We obtained our isochrones from Girardi et al. 2026 using the PARSEC tracks from Bressan et al. 2012.

We know that the Monkey Head Galaxy sits at approximately 2.2 kpc with an age of 2 Myr (Khachikian et al. 2005). Given this, we would assume the cluster to be dominated by main sequence and pre-main sequence stars. As such, we should fit our data such that it sits over these respective stellar phases in our isochrones. See Figure 3.

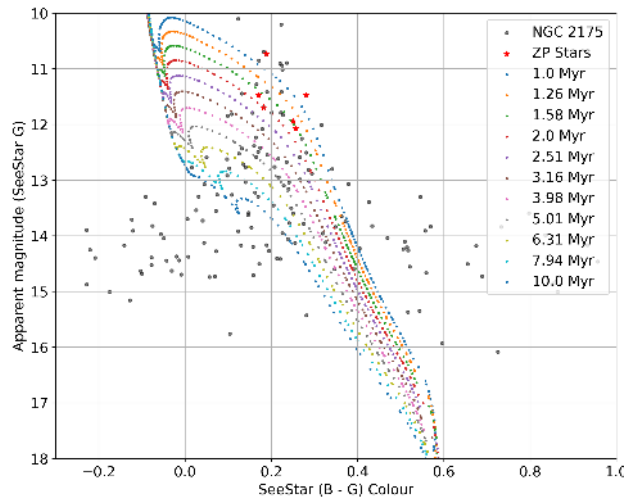


Figure 3: Sets of Girardi isochrones overlaid on our SeeStar data.

The distance modulus used to obtain this fit was 11.1. The distance modulus is

$$d = 10^{(\text{modulus}+5)/5}. \quad (6)$$

Using Equation 6, the calculated distance for NGC 2175 is 1660 parsecs. There are some visual artefacts that we can see in the figure that are of interest. Notably, there is quite a large colour spread in our data. This suggests that a lot of the captured objects don't lie within our object of interest. When we filtered out our data, we gave an optimistic estimate for the pixel radius of our cluster in hopes to mitigate background stars, however this effect only works laterally, not for objects directly in the foreground or background of the cluster. This spread is, thus, not a concern as we can merely keep our focus on the pre-main/main sequence stars in the central part of our spread. This is defined by the other notable feature of the plot, which is the location of our calibration stars. We can see that we have successfully chosen stars with reasonably bright magnitudes and their locations on the diagram are expected.

From this fit we observe a main sequence in our data that approximately aligns with an age of 2 to 3.16 Myr based on the isochrones used. This is a reasonable range to estimate for the age of the cluster.

## 4 Discussion

Our measurement of 1660 parsecs falls short of the literature (Khachikian et al. 2005) by a distance modulus of  $\approx 0.6$  magnitudes, representing a narrow difference. Despite our close result, there are some notable effects that would produce an inaccurate result.

Since NGC 2175 is a very young cluster, it has a lot of low mass stars still in the pre-main sequence. This means they haven't yet finished collapsing into stable main sequence stars. As a result, they would have a larger radius than a main sequence star of the same temperature, appearing intrinsically brighter than the isochrones can predict. Since we are trying to achieve a fit with our data to the isochrones, it is highly likely that our data has been fit too bright on the isochrones since the code must assume in such a case that the stars are brighter because they are closer, not because they aren't on the main sequence. This decreases the distance modulus, underestimating the distance. Another significant effect that would further underestimate the distance is having unresolved binary stars. It is typical for over 30% of stellar objects in young open clusters to be binary systems, appearing  $\approx 0.75$  magnitudes brighter than if they were two separate stars (Hurley and Tout 1998). Since they appear brighter, we would need to decrease the distance modulus to compensate and thus, decrease the measured distance. The distance modulus itself encapsulates the effect of interstellar dust (reddening). Since we know the monkey head nebula is an active region with lots of interstellar dust, the local extinction would vary across the image, warping our observations. It is hard to decouple the effects of interstellar dust and we did not account for this, meaning some stars might appear dimmer, increasing our distance modulus slightly.

As for the age estimate, it was highly difficult to determine an appropriate age window but it was very clear that the literature of 2 Myr lies within the range of acceptable values for the age. The determined window of 2-3.16 Myr is a representation based on an assumption of a simple stellar population that in reality is unlikely to be the case. The simple stellar population approximation assumes that all the stars were born at the same time, however since this is not the case, the best we can do is a window determined by our isochrone fitting. We determined that the main sequence that approximately runs through our calibration stars and the centroid of the data (ignoring most of the foreground/background spread) aligns within the 2-3.16 Myr bracket.

A limitation encountered here is the resolution of the isochrone dataset. We cannot resolve more detailed age isochrones in the database, so the safest measurement is to take a small window as the age of the cluster. Given this, a result of 2-3.16 Myr for the age of NGC 2175 is a sound result that closely aligns with literature.

This age we found and the age from literature is quite a young open cluster. Open clusters like this tend to full of gas and dust interacting and fueling star formation. Young open clusters tend to have high rates of star formation which is ideal feature to explore the evolution of stars and the development of clusters.

In Massi et al. 2015, the young open cluster NGC 6357 was used to investigate a young stellar populations. For this cluster, data from public and private sources were used to carefully determine properties of the young stars. In it, the spatial distribution and substructures of the stellar clusters present were able to be identified near the region. Furthermore, the ages of stars and the age distributions was determined for the cluster. This study was done to help us further understand the process of star formation, what can trigger it and how star cluster may come to form.

NGC 2175 is quite a similar distance away at approximately 1.66kpc away while NGC 6357 is 1.7kpc away. NGC 2175 is also in a similar age range of 2.0 to 3.16 Myr compared to NGC 6357 being approximately 1 to 3 Myr (Massi et al. 2015). Since these clusters both have these similar properties, they are a good comparison to find differences in the stellar evolution. An issue that Massi et al. 2015 had was that NGC 6357 was close to the galactic plane which complicated the processing steps. This is also an issue with NGC 2175 which must be addressed.

Overall, NGC 2175 is a great candidate to analyse young stellar populations. In our findings, we were able to identify 398 celestial objects within the cluster. This gives a great baseline to analyse the stellar properties and overall distribution of the properties.

The analysis of star formation within the Monkey Head Nebula which encapsulates NGC 2175 can be seen in Chibueze et al. 2013. This shows successful results in identifying the causality of star formation. This can be pertinent to analysing stellar distributions in young open clusters. Chibueze et al. 2013 showed that local star formation in a region of the nebula was spontaneous rather than sequentially caused by other stars. This information can be used in the potential analysis of stellar distributions in NGC 2175.

## 5 Conclusion

This observational study follows the creation of a computational photometric process whereby a consumer grade SeeStar image of the young open cluster NGC 2175 was analysed. The individual sources in the sky were isolated making use of the Photutils library and cross matched to the Gaia catalogue. By mapping the data onto theoretical Girardi isochrones we determined the core cluster features. The observed age was measured to be 2-3.16 Myr, which aligns with the standard literature value of 2 Myr. The observed physical distance was 1660 pc, which lies short of the 2200 standard literature value, yet we determined that due to the logarithmic nature of the magnitude, this is an impressive result for consumer grade photometry (within 0.6 magnitudes). The factors affecting these results were discussed, including the presence of a high percentage of pre-main sequence stars and unresolved binaries, among other sources of systematic errors. The study proves that high-quality photometry can be performed without high-fidelity commercial equipment.

## 6 Acknowledgements

This work made use of Jupyter, astropy, scipy, numpy, ccdproc, astroalign, AstrometryNet, and photutils software. We made use of code examples provided in the ASP3231 resourcesBrown 2026. Assistance and feedback provided by Michael Brown was helpful in the development of our data processing.

## 7 References

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## A Processed images in the green and blue filters

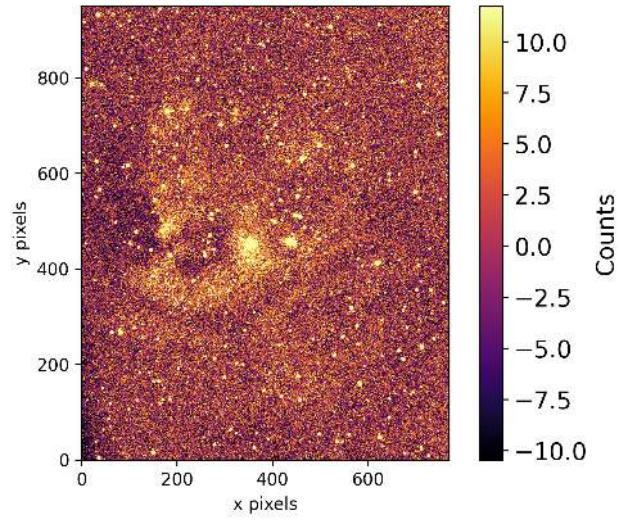


Figure 4: Final processed image of NGC 2175 in the green filter.

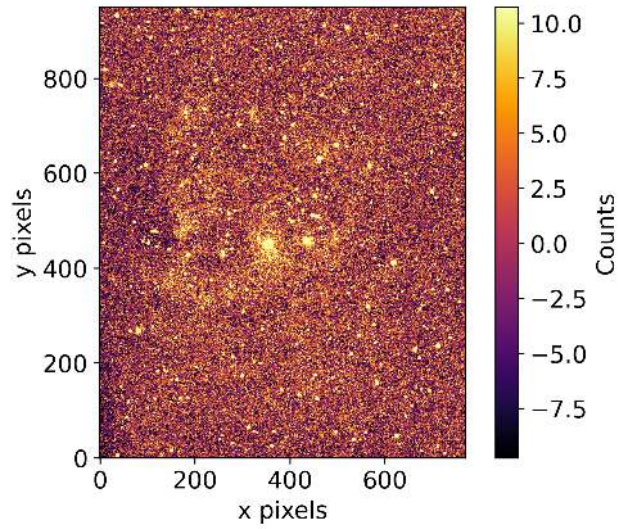


Figure 5: Final processed image of NGC 2175 in the blue filter.

## B Stars in our catalogue of NGC 2175

Table 1: Top 10 Brightest Photometric Sources Detected in NGC 2175

| ID  | $X_{\text{pixel}}$ | $Y_{\text{pixel}}$ | $f$       | $R_{\text{SeeStar}}$ | $G_{\text{SeeStar}}$ | $B_{\text{SeeStar}}$ | RA (°) | Dec (°) |
|-----|--------------------|--------------------|-----------|----------------------|----------------------|----------------------|--------|---------|
| 175 | 355.88             | 450.56             | 259894.04 | 7.58                 | 7.65                 | 7.86                 | 92.41  | 20.49   |
| 288 | 461.69             | 630.00             | 27545.60  | 10.00                | 10.10                | 10.23                | 92.46  | 20.62   |
| 92  | 82.02              | 266.66             | 24468.19  | 10.13                | 10.21                | 10.43                | 92.26  | 20.33   |
| 250 | 452.51             | 550.75             | 14640.05  | 10.42                | 10.70                | 10.88                | 92.47  | 20.57   |
| 173 | 185.72             | 427.82             | 14400.26  | 10.60                | 10.73                | 10.92                | 92.30  | 20.45   |
| 283 | 97.21              | 616.45             | 12381.86  | 10.50                | 10.85                | 11.02                | 92.21  | 20.56   |
| 328 | 181.10             | 728.60             | 12604.23  | 10.46                | 10.90                | 11.13                | 92.25  | 20.64   |
| 191 | 437.14             | 457.50             | 19040.76  | 10.47                | 10.90                | 11.12                | 92.47  | 20.50   |
| 232 | 453.14             | 509.43             | 29127.16  | 10.86                | 10.94                | 11.17                | 92.47  | 20.54   |
| 171 | 259.45             | 427.68             | 12115.33  | 10.18                | 11.02                | 11.24                | 92.35  | 20.46   |